SAT MATH

Practice tests • Practice Test 1 • Module 2, easier (22 questions, 35 minutes) • Answer key

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Answer key with worked solutions

Directions

These questions follow the format of the SAT Math section. A calculator is allowed on every question (Desmos is built into the digital SAT). Multiple-choice questions have one best answer. For **student-produced responses**, write your answer on the line: a fraction or a decimal is accepted (for example $3/4$ or 0.75); give negative answers with a minus sign; do not use mixed numbers.

Question 1 (Algebra)

If $x + 9 = 15$, what is the value of x?

(A) 24

(B) 6

(C) -6

(D) 9

Solution

Answer: (B). $x = 15 - 9 = 6$.

Question 2 (Algebra)

If $4x = 28$, what is the value of x?

(A) 7

(B) 24

(C) 32

(D) 112

Solution

Answer: (A). $28 \div 4 = 7$.

Question 3 (Data Analysis)

Three notebooks cost $7.50. At the same price, how much do five notebooks cost?

(A) $10.00

(B) $15.00

(C) $12.50

(D) $37.50

Solution

Answer: (C). One notebook costs $2.50; five cost $12.50.

Question 4 (Advanced Math)

Which expression is equivalent to $3(x^2 + 2x) - x^2$?

(A) $2x^2 + 2x$
(B) $3x^2 + 6x$
(C) $4x^2 + 6x$
(D) $2x^2 + 6x$

Solution
Answer: (D). $3x^2 + 6x - x^2 = 2x^2 + 6x$.

Question 5 (Algebra) STUDENT-PRODUCED RESPONSE

The function f is defined by $f(x) = 2x + 7$. What is the value of $f(5)$?

Solution
Answer: 17. $2(5) + 7 = 17$.

Question 6 (Geometry and Trig)

A square has a perimeter of 36. What is its area?

(A) 36
(B) 81
(C) 144
(D) 9

Solution
Answer: (B). Each side is 9; the area is 81.

Question 7 (Data Analysis)

What is the median of $2,\ 9,\ 4,\ 7,\ 5$?

(A) 4
(B) 5.4
(C) 7
(D) 5

Solution
Answer: (D). In order: $2, 4, 5, 7, 9$. The middle value is 5.

Question 8 (Algebra)

What is the slope of the line $y = -3x + 4$?

(A) -3
(B) 4
(C) 3
(D) $\frac{4}{3}$

Solution
Answer: (A). In $y = mx + b$, $m = -3$.

Question 9 (Advanced Math)

If $x^2 = 36$ and $x > 0$, what is the value of x?

(A) 18
(B) 72
(C) 6
(D) -6

Solution
Answer: (C). $x = \sqrt{36} = 6$.

Question 10 (Algebra) STUDENT-PRODUCED RESPONSE

What is the solution to $2x + 3 = x + 10$?

Solution
Answer: 7. $x = 10 - 3 = 7$.

Question 11 (Geometry and Trig)

In a right triangle, one acute angle measures $35°$. What is the measure of the other acute angle?

(A) $145°$
(B) $55°$
(C) $65°$
(D) $45°$

Solution
Answer: (B). $180 - 90 - 35 = 55$.

Question 12 (Data Analysis)

What is 25% of 80?

(A) 25
(B) 32
(C) 55
(D) 20

Solution
Answer: (D). $\frac{1}{4}(80) = 20$.

Question 13 (Advanced Math)

The function f is defined by $f(x) = x^2 + 1$. What is the value of $f(3)$?

(A) 10
(B) 7
(C) 9
(D) 16

Solution
Answer: (A). $9 + 1 = 10$.

Question 14 (Algebra)

A phone plan costs \$15 per month plus \$0.10 per text message. What is the cost of a month with 120 text messages?

(A) \$12
(B) \$135
(C) \$27
(D) \$16.20

Solution
Answer: (C). $15 + 0.10(120) = 27$.

Question 15 (Advanced Math) STUDENT-PRODUCED RESPONSE

What is the positive solution of $(x-2)(x+5) = 0$?

Solution
Answer: 2. The solutions are 2 and -5.

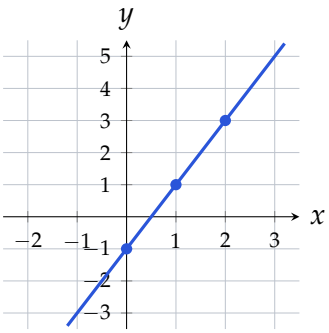
Question 16 (Geometry and Trig) STUDENT-PRODUCED RESPONSE

A circle has diameter 10. Its circumference is $k\pi$. What is the value of k?

Solution
Answer: 10. $C = \pi d = 10\pi$.

Question 17 (Algebra)

Which equation represents the line shown?



(A) $y = x - 1$
(B) $y = -2x - 1$
(C) $y = 2x + 1$
(D) $y = 2x - 1$

Solution
Answer: (D). Through $(0, -1)$ and $(1, 1)$: slope 2, intercept -1.

Question 18 (Data Analysis)

A bag holds 3 red, 5 blue and 2 green marbles. One marble is drawn at random. What is the probability that it is blue?

(A) $\frac{1}{2}$
(B) $\frac{1}{5}$
(C) $\frac{5}{3}$
(D) $\frac{3}{10}$

Solution
Answer: (A). $\frac{5}{10} = \frac{1}{2}$.

Question 19 (Advanced Math)

Which expression is equivalent to $x^3 \cdot x^4$?

(A) x^{12}
(B) x^7
(C) $2x^7$
(D) x^1

Solution
Answer: (B). Add the exponents: $x^{3+4} = x^7$.

Question 20 (Algebra) STUDENT-PRODUCED RESPONSE

If $x + y = 12$ and $x - y = 2$, what is the value of y?

Solution
Answer: 5. Subtract: $2y = 10$, $y = 5$.

Question 21 (Geometry and Trig)

The legs of a right triangle are 6 and 8. What is the length of the hypotenuse?

(A) 14
(B) $\sqrt{28}$
(C) 10
(D) 48

Solution
Answer: (C). $\sqrt{36 + 64} = 10$.

Question 22 (Advanced Math) STUDENT-PRODUCED RESPONSE

If $2^x = 64$, what is the value of x?

Solution

Answer: 6. $64 = 2^6$.